



USN

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations
 December 2025
MA2003-1 – VECTOR CALCULUS AND COMPLEX FUNCTIONS
 (For CV, ME)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- In spherical coordinate system (r, θ, ϕ) , $r = a$, where a is a constant represents
 - A plane
 - A Sphere with radius a
 - A cone
 - A cylinder with radius a
- A unit tangent vector to the surface $x = t, y = t^2, z = t^3$ at $t = 1$ is
 - $\frac{\hat{i} + 2\hat{j} + 3\hat{k}}{\sqrt{14}}$
 - $\frac{\hat{i} - 2\hat{j} + 3\hat{k}}{\sqrt{14}}$
 - $\frac{2\hat{i} + \hat{j} + \hat{k}}{\sqrt{6}}$
 - $\frac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}}$
- $\text{div}(\text{curl } \vec{F}) =$
 - 1
 - 2
 - $\vec{0}$
 - 0
- $\vec{F} = yz\hat{i} + zx\hat{j} + xy\hat{k}$ is
 - Irrotational vector field
 - Solenoidal vector field
 - Both solenoidal and irrotational vector field
 - $\nabla \cdot \vec{F} = 5$
- If $\vec{F}$ is conservative force field then $\text{curl } \vec{F}$ is
 - $\nabla \cdot \vec{F}$
 - $\vec{F}$
 - $\vec{0}$
 - 0
- Using Green's theorem in the plane, the value of the integral $\frac{1}{2} \int_C xdy - ydx$ is
 - Area of the region R bounded by C
 - 1
 - 1/2
 - 0
- Gauss divergence theorem relates
 - Line and surface integral
 - Surface and volume integral
 - Line and volume integral
 - Line integral, surface integral and volume integral
- The line integral $\int_C \vec{F} \cdot d\vec{r}$ represents:
 - Area under the curve
 - Flux through the surface
 - Volume generated by rotation
 - Work done by the vector field $\vec{F}$ along path C
- If $f(z) = u(x, y) + iv(x, y)$ is analytic, then $f'(z) =$
 - $\frac{\partial u}{\partial x} - i \frac{\partial u}{\partial y}$
 - $\frac{\partial u}{\partial x} - i \frac{\partial v}{\partial x}$
 - $\frac{\partial v}{\partial y} - i \frac{\partial v}{\partial x}$
 - $\frac{\partial u}{\partial y} - i \frac{\partial v}{\partial x}$

10. If $u = 2x - x^2 + ay^2$ is to be harmonic, then a should be
 A) 0 B) 2
 C) 3 D) 1
11. If the real part of an analytic function is $u(x, y) = x^2 - y^2$, then using Milne-Thomson method, the analytic function $f(z)$ is:
 A) $f(z) = \bar{z}^2$ B) $f(z) = z^2$
 C) $f(z) = 2xy + i(x^2 - y^2)$ D) $f(z) = x^2 + y^2 + i 2xy$
12. The bilinear transformation $w = \frac{2z+3}{z+2}$ maps $z = -2$ to:
 A) $w = 0$ B) $w = -1$
 C) $w = \infty$ D) $w = 1$
13. The value of $\int_{|z|=2} \frac{dz}{z}$ is:
 A) 0 B) πi
 C) $-2\pi i$ D) $2\pi i$
14. Cauchy's integral formula states that if $f(z)$ is analytic inside and on a simple closed contour C and a is inside C , then:
 A) $f'(a) = \frac{1}{2\pi i} \int_C f(z) dz$ B) $f(a) = \frac{1}{2\pi i} \int_C \frac{f(z)}{z-a} dz$
 C) $f(a) = \frac{1}{2\pi i} \int_C f(z) dz$ D) $f(a) = \int_C f(z) dz$
15. If $f(z) = \frac{10+6z^2}{(z-1)(z+2)(z-3)^3}$ then $z = 3$ is a pole of $f(z)$ of order _____
 A) 3 B) 1
 C) 2 D) 5
16. A Laurent series of $f(z)$ around $z = z_0$ can be written as:
 A) $\sum_{n=0}^{\infty} a_n (z - z_0)^n$ B) $\sum_{n=0}^{\infty} a_n e^{n(z-z_0)}$
 C) $\sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$ D) $\sum a_n \cos n (z - z_0)$
17. If $f(x)$ is even, its Fourier series contains:
 A) Only sine terms B) Constant only
 C) Both sine and cosine terms D) Only cosine terms
18. The Fourier series of $f(x) = x$ on $[-\pi, \pi]$ contains:
 A) Only sine terms B) Only cosine terms
 C) Both sine and cosine terms D) Constant term only
19. If $F\{f(t)\} = F(\omega)$, then $f(t - a) =$ _____
 A) $F(\omega)e^{-i\omega a}$ B) $F(\omega)e^{i\omega a}$
 C) $F(\omega - a)$ D) $F(\omega + a)$
20. Fourier transform of $f(t) = e^{-|t|}$ is:
 A) $\frac{1}{1 + \omega^2}$ B) $\frac{2}{1 + \omega^2}$
 C) $\frac{1}{\omega}$ D) $\frac{1}{\omega^2}$

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit - I

- | | Marks | BT* | CO* | PO* |
|---|-------|-----|-----|-----|
| 1. a) Determine the magnitude of velocity and acceleration at $t = 0$ of a particle moving along a curve whose parametric equations are $x = e^{-t}$, $y = 2 \cos 3t$, $z = 2 \sin 3t$, where t is the time. | 6 | L*3 | 1 | 2 |
| b) Prove that cylindrical coordinate system is orthogonal curvilinear coordinate system. | 6 | L2 | 1 | 1 |
| c) Determine whether the following force field is conservative or not.
$\vec{F} = 2xz \hat{i} + (x^2 - y) \hat{j} + (2z - x^2) \hat{k}$ | 4 | L2 | 2 | 1 |

2. a) Find the total work done in moving a particle in a force field given by $\vec{F} = 3xy\hat{i} - 5z\hat{j} + 10x\hat{k}$ along the curve $x = t^2 + 1, y = 2t^2, z = t^3$ from $t = 1$ to $t = 2$. 6 L3 2 2
- b) State and prove the Green's theorem. 6 L2 2 1
- c) What is the greatest rate of increase of $u = x^2 + yz^2$ at the point $(1, -1, 3)$? 4 L2 1 1
3. a) Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at $(1, 2, 2)$. 6 L2 1 1
- b) Using Stoke's theorem evaluate $\int_C \vec{F} \cdot d\vec{r}$ for the vector field $\vec{F} = (x^2 + y^2)\hat{i} - 2xy\hat{j}$ round the rectangle in the plane $z = 0$ and bounded by the lines $x = 0, y = 0, x = a$ and $y = b$. 6 L3 2 2
- c) Prove that $\nabla \cdot (\nabla \times \vec{F}) = 0$. 4 L2 1 1

Unit – II

4. a) Show that $v = e^x \sin y$ is harmonic. Find u such that $f(z) = u + iv$ is analytic. 6 L2 3 1
- b) If $f(z) = u(x, y) + i v(x, y)$ is analytic in a region R , then prove that $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ & $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$. 6 L2 3 1
- c) Evaluate $\int_C \frac{dz}{z^2 - z}$, if $C: |z| = \frac{1}{2}$ using the Cauchy's integral formula. 4 L3 4 2
5. a) Evaluate $\int_C \frac{z^2}{(z-1)^2(z+2)} dz$ where C is the circle $|z| = \frac{5}{2}$ using the residue theorem. 6 L3 4 2
- b) Write the Laurent's series expansion of $\frac{1}{z^2 - 3z + 2}$ in the region $|z| > 2$. 6 L2 4 1
- c) Show that $w = \log z$ is analytic. 4 L2 3 1
6. a) Evaluate the bilinear transformation which maps $z = 0, 1, \infty$ onto $w = -1, -i, 1$. 6 L3 3 1
- b) State and prove the Cauchy's integral theorem. 6 L2 4 1
- c) Find the poles and residues at each pole of the function $f(z) = \frac{2z+1}{z^2 - z - 2}$. 4 L2 4 2

Unit – III

7. a) Evaluate a_0, a_1, b_1 of the Fourier series of the following data.

x	0	$\pi/3$	$2\pi/3$	π	$4\pi/3$	$5\pi/3$	2π
y	1	1.4	1.9	1.7	1.5	1.2	1

- b) Solve the Fourier series of $f(x) = x^2$ in $(-\pi, \pi)$ with $f(x + 2\pi) = f(x)$. 6 L3 5 1
- c) Solve the half range Fourier sine series of $f(x) = x$ in $0 < x < \pi$. 6 L2 5 2
8. a) State and prove the convolution theorem of Fourier transforms. 4 L2 5 1
- b) Evaluate the Fourier transform of $e^{-a|t|}, t \geq 0, a > 0$. 6 L3 5 2
- c) Show that $F_c\{t f(t)\} = \frac{d}{d\omega}\{F_s(\omega)\}$. 4 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
